



1. Solving homogeneous systems of linear ODEs

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Objectives

- Describe systems of ODEs using linear algebra notation.
- Find a basis for the vector space of homogenous solutions in terms of eigenvalues and eigenvectors of the associated matrix.
- Express solutions to homogenous systems of ODEs in terms of a **fundamental matrix**.
- Find a **companion matrix** expressing a collection of scalar ODEs as a single system of first order ODEs.
- Find the system of ODEs that describes the dynamics of a **tuned mass damper** and **coupled oscillator**.

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